

Member 2 – Pose Detection Accuracy Analysis Responsibilities

AI/ML Internship — Day 16

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1. Introduction

Pose Detection is a computer vision technique that identifies and tracks key human body landmarks from images and videos. It serves as a foundational technology for Virtual Try-On systems, fitness tracking, human activity recognition, motion capture, augmented reality, healthcare monitoring, and avatar generation.

Modern AI-powered pose estimation models use deep learning and computer vision algorithms to accurately locate body joints and skeletal structures. These landmarks help systems understand human posture and movement.

For Virtual Try-On applications, pose detection is essential because garment alignment depends heavily on accurate body landmark localization.

This research evaluates pose estimation models under different conditions and analyses their performance for Virtual Try-On systems.

2. Objectives

The objectives of this study are:

- Understand pose estimation techniques.
 - Evaluate body landmark detection accuracy.
 - Compare MediaPipe Pose and OpenPose.
 - Analyse performance under different body poses.
 - Study lighting effects on detection accuracy.
 - Determine suitability for Virtual Try-On applications.
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3. Understanding Human Pose Estimation

Human pose estimation refers to the process of detecting body joints and skeletal structures from visual data.

The output typically consists of landmark coordinates representing:

- Head
- Neck
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

These landmarks enable AI systems to reconstruct human posture and movement.

4. MediaPipe Pose

Overview

MediaPipe Pose is a lightweight real-time pose estimation framework developed by Google.

It detects 33 body landmarks and is optimized for mobile, desktop, and web applications.

Features

- Real-time pose estimation
- 33 body landmarks
- Lightweight architecture
- Cross-platform deployment
- Fast processing speed

Advantages

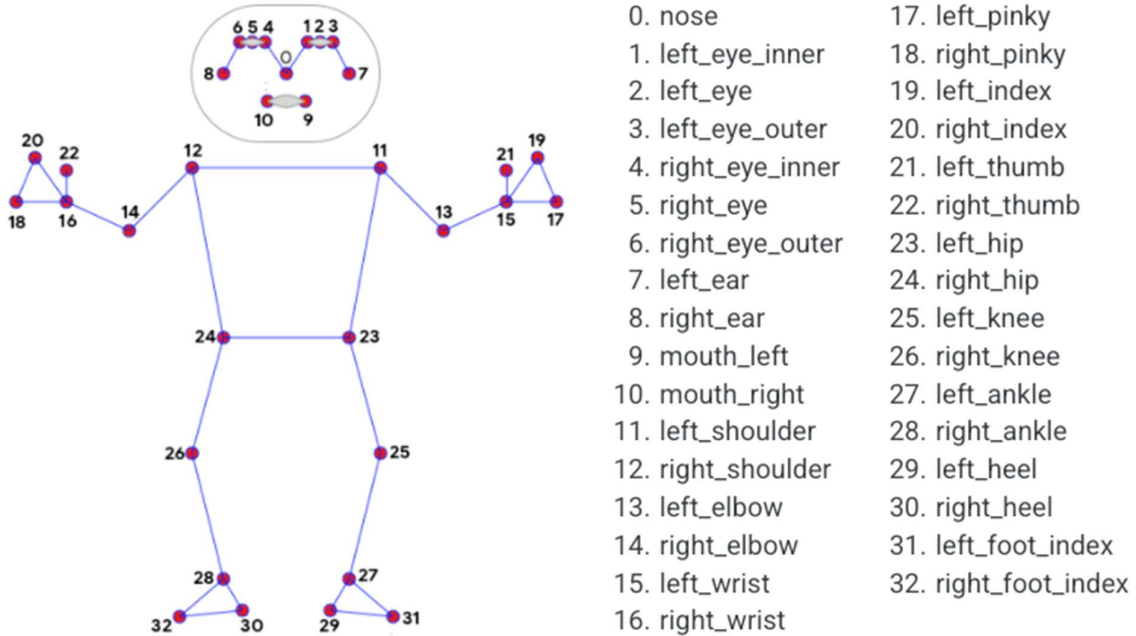
- High inference speed
- Low computational requirements
- Easy implementation
- Mobile-friendly

Limitations

- Reduced accuracy under extreme poses
- Performance affected by severe occlusions
- Less robust than OpenPose in multi-person scenarios

Applications

- Fitness tracking
- Virtual Try-On systems
- Gesture recognition
- AR experiences
- Motion analysis



5. OpenPose

Overview

OpenPose is a deep-learning-based pose estimation framework capable of detecting full-body, hand, foot, and facial landmarks.

It is widely used in research and industrial applications.

Features

- Multi-person detection
- Body landmark tracking
- Hand key point detection
- Facial landmark detection
- High precision estimation

Advantages

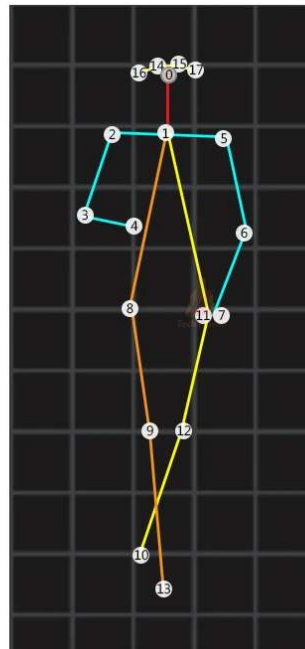
- Excellent accuracy
- Robust landmark localization
- Multi-person support
- Detailed body tracking

Limitations

- Higher computational cost
- Requires powerful hardware
- Slower inference speed

Applications

- Motion capture
- Sports analytics
- Healthcare systems
- Human-computer interaction
- Virtual Try-On systems



6. Experimental Setup

The pose estimation models were evaluated using images under different conditions.

Test Scenarios

Front View

The subject faces directly toward the camera.

Side View

The subject rotates approximately 90 degrees.

Arms Raised

Both arms are positioned above shoulder level.

Walking Pose

Dynamic body movement simulation.

Low Lighting

Reduced illumination environment.

Complex Background

Presence of distracting background elements.

7. Landmark Detection Analysis

The following landmarks were analysed:

Landmark	Purpose
Shoulder	Garment alignment
Elbow	Arm tracking
Wrist	Sleeve fitting
Hip	Lower body estimation
Knee	Pose understanding
Ankle	Full body tracking

8. MediaPipe Pose Results

Front View

Performance: Excellent

Observations:

- Accurate shoulder detection
- Stable landmark positioning
- Fast inference speed

Side View

Performance: Good

Observations:

- Slight shoulder landmark displacement
- Reduced visibility of hidden joints

Arms Raised

Performance: Good

Observations:

- Stable arm tracking
- Minor wrist fluctuations

Low Lighting

Performance: Moderate

Observations:

- Reduced confidence scores
- Occasional landmark instability

Complex Background

Performance: Good

Observations:

- Most landmarks remained stable
- Minor detection noise



9. OpenPose Results

Front View

Performance: Excellent

Observations:

- Accurate landmark placement
- Stable skeleton generation

Side View

Performance: Very Good

Observations:

- Better side-body tracking than MediaPipe
- Improved hidden joint estimation

Arms Raised

Performance: Excellent

Observations:

- Consistent wrist detection
- Stable arm tracking

Low Lighting

Performance: Good

Observations:

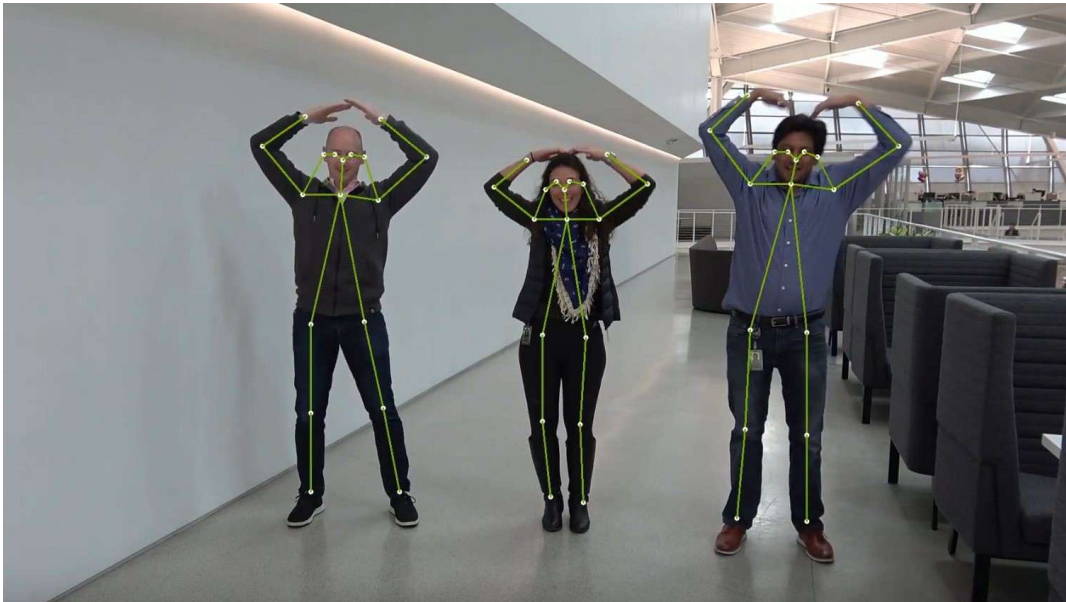
- Better robustness than MediaPipe
- Reduced sensitivity to illumination changes

Complex Background

Performance: Very Good

Observations:

- Better landmark separation
- Improved background handling



10. Accuracy Comparison

Criteria	MediaPipe Pose	OpenPose
Detection Speed	Excellent	Good
Accuracy	Good	Excellent
Side View Tracking	Good	Very Good
Multi-Person Detection	Limited	Excellent
Low-Light Performance	Moderate	Good
Hardware Requirements	Low	High
Mobile Deployment	Excellent	Moderate
Virtual Try-On Suitability	Excellent	Excellent

11. Performance Evaluation

Detection Speed

Best Model

MediaPipe Pose

Reason:

- Lightweight architecture
 - Real-time processing
 - Mobile optimization
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Landmark Accuracy

Best Model

OpenPose

Reason:

- More detailed body modelling
 - Better landmark precision
 - Robust body tracking
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Virtual Try-On Applications

Recommended Model

MediaPipe Pose

Reason:

- Faster processing
 - Suitable for real-time fitting
 - Lower hardware requirements
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Research Applications

Recommended Model

OpenPose

Reason:

- Higher landmark precision
 - Better tracking quality
 - Advanced pose estimation
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12. Challenges Identified

Several challenges were observed during testing.

Occlusion Problems

Body parts become hidden behind other objects or body regions.

Lighting Variations

Low-light conditions reduce detection confidence.

Side View Detection

Partial body visibility affects landmark estimation.

Motion Blur

Fast movement introduces tracking instability.

Background Complexity

Busy scenes may introduce detection noise.

13. Recommendations

For Virtual Try-On systems:

MediaPipe Pose

Recommended for:

- Real-time systems
- Mobile applications
- AR environments

OpenPose

Recommended for:

- Research projects
- High-accuracy applications
- Advanced body analysis

14. Future Improvements

Potential improvements include:

- Multi-view pose estimation
- 3D landmark reconstruction
- Better occlusion handling
- Improved low-light detection
- Hybrid pose estimation systems
- AI-based landmark refinement

15. Learning Outcomes

During this research, the following concepts were studied:

- Human pose estimation
- Landmark detection
- Skeletal tracking
- Computer vision workflows
- Deep learning pose models
- Real-time body tracking
- Virtual Try-On integration
- Performance evaluation methods

16. Conclusion

This study evaluated MediaPipe Pose and OpenPose under different environmental and body pose conditions.

The findings indicate that MediaPipe Pose provides excellent speed and deployment flexibility, making it suitable for real-time Virtual Try-On systems. OpenPose delivers superior landmark accuracy and robustness, making it suitable for research and high-precision applications.

Both technologies play a critical role in Virtual Try-On pipelines and future AI-powered fashion systems. The selection of a model depends on the balance between speed, accuracy, and hardware availability.